perplexity

## Principle (Simple Version)

This experiment uses **potassium permanganate ($KMnO_4$)** as an oxidizing agent.

- **$KMnO_4$ + acid** → oxygen released.
- The oxygen oxidizes **ferrous sulphate ($FeSO_4$)** into ferric sulphate ($Fe_2(SO_4)_3$).
- The oxygen can also oxidize **oxalic acid ($H_2C_2O_4$)** into carbon dioxide and water.

**$KMnO_4$ acts as its own indicator**: when the reaction finishes, the first permanent pale pink color means the end point.

This is done in two parts:

1. Standardize $KMnO_4$ using a standard **Mohr's salt solution**.
2. Use the standardized $KMnO_4$ to find the strength and amount of the given oxalic acid solution.

## Procedure (Easy Steps)

1. **Prepare Standard Mohr's Salt Solution:**
   - Weigh 3.92 g crystals of Mohr's salt.
   - Dissolve in water with 40 ml dilute sulphuric acid (to stop hydrolysis).
   - Make total volume 100 ml.
2. **Standardize $KMnO_4$:**
   - Pipette 20 ml of Mohr's salt solution.
   - Add the same volume of dilute sulphuric acid.
   - Titrate with $KMnO_4$ in the burette until the first permanent pale pink color appears.
   - Repeat until results are close (concordant).
3. **Titrate Oxalic Acid:**
   - Pipette 20 ml of the given oxalic acid solution.
   - Add 20 ml sulphuric acid and warm the mixture gently.
   - Titrate with the standardized $KMnO_4$ until pale pink color appears.
   - Repeat for concordant readings.
4. **Calculation:**

- Use titre values from $KMnO_4$ to determine the concentration and mass of the oxalic acid in the solution.